

# Empty Shelf-Detection Model

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**Abstract**—Retail inventory management faces challenges in identifying empty shelf spaces in real time, as manual inspection is often slow and prone to errors. ShelfSight is a computer vision-based solution that uses YOLOv8 to automatically detect empty shelf regions from live camera feeds or recorded video. The system highlights unoccupied spaces using bounding boxes, enabling store personnel to quickly identify restocking needs. Developed using a pre-trained model and a Roboflow dataset, ShelfSight provides a cost-effective and scalable solution for automated shelf monitoring. It improves operational efficiency, reduces manual effort, and supports timely inventory management. Future enhancements include improving detection accuracy, integrating with inventory systems, and enabling advanced analytics for retail operations.

## I. INTRODUCTION

In the modern retail ecosystem, efficient inventory management and product availability are critical for customer satisfaction and revenue generation. A major challenge faced by retailers is the occurrence of out-of-stock (OOS) situations, where products are unavailable on shelves despite existing demand. These empty shelf spaces lead to lost sales, reduced customer trust, and weakened brand loyalty. Studies indicate that OOS rates typically range between 5% and 15%, resulting in significant financial losses each year.

Traditionally, retailers rely on manual shelf inspection, where staff physically check aisles to identify stock issues. Although simple to implement, this method is time-consuming, labor-intensive, and prone to human error, delays, and inconsistencies. In large retail environments with thousands of products, manual monitoring becomes inefficient and limits the responsiveness of inventory management systems.

With advancements in Deep Learning and Computer Vision, automated visual inspection has emerged as a practical alternative. Techniques such as Convolutional Neural Networks (CNNs) and object detection models like YOLO (You Only Look Once) enable real-time detection and localization of objects in images and videos. These technologies allow con-

tinuous monitoring of shelves with improved accuracy and minimal human intervention.

However, many existing solutions face challenges such as high deployment cost, computational requirements, and limited adaptability to different retail environments. These limitations highlight the need for scalable, efficient, and cost-effective systems that can operate reliably under varying conditions.

To address these challenges, the proposed system, *ShelfSight*, utilizes YOLOv8 for automatic detection of empty shelf spaces. The system processes live camera feeds as well as recorded video data to identify empty regions and highlight them using bounding boxes. This enables timely restocking and improves overall operational efficiency.

The system is trained and validated on a publicly available dataset, ensuring reliable performance across different shelf layouts, lighting conditions, and product arrangements. The implementation is based on an open-source GitHub repository, allowing easy setup, reproducibility, and real-time deployment using pre-trained models and standard libraries.

In addition, the proposed system improves store efficiency by reducing the time required to identify and resolve stock issues. Continuous monitoring minimizes dependency on manual inspection and enables faster response to empty shelf conditions. The use of an open-source framework also makes the system cost-effective and suitable for deployment in various retail environments.

Furthermore, the system can be extended with advanced features such as automated alert generation, integration with inventory management systems, and data analytics for demand forecasting. These enhancements can support better decision-making and resource optimization. Overall, *ShelfSight* provides a practical and scalable approach to intelligent retail automation, effectively addressing the critical issue of product availability.

## II. RELATED WORK

The challenge of automated shelf monitoring has been addressed by various studies focusing on improving accuracy, efficiency, and real-time performance. Traditional manual inspection methods rely on employees physically walking through aisles, which is time-consuming, labor-intensive, and prone to human error.

Several computer vision approaches have been developed to address this gap. Early attempts employed traditional image processing techniques such as edge detection and histogram analysis. Tonioni et al. [1] proposed a method using color histograms and texture analysis to identify empty spaces. However, these approaches were sensitive to lighting variations and occlusions.

The introduction of deep learning significantly improved object detection performance. Models such as Faster R-CNN, SSD (Single Shot Detector), and YOLO have been widely applied in retail scenarios. Varol et al. [2] used Faster R-CNN for shelf product detection, achieving high accuracy but

with slower inference speeds. George and Rout [3] compared SSD and YOLOv3, concluding that YOLOv3 provides a better balance between accuracy and real-time performance.

Location-based and planogram compliance methods have been explored in works like Marder et al. [4], who developed a system combining object detection with planogram alignment to detect missing items. However, this required precise shelf layout information and camera calibration.

Recent advancements have explored two-stage approaches involving product detection followed by gap analysis. Liu et al. [5] introduced this method to identify empty spaces, though it involved additional post-processing complexity.

Despite these developments, there remains a need for a lightweight, real-time solution that can detect empty shelf spaces without requiring extensive computational resources or precise calibration. The proposed ShelfSight application builds on these concepts by leveraging YOLOv8's anchor-free detection and efficient architecture within a deployable framework. Table I presents a summary of related research and methodologies relevant to empty shelf detection.

TABLE I: Literature Review

| Year | Author                                                     | Title                                                                                           | Methodology                                                                                                                                                                          |
|------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Dipendra Jha, Ankit R. Khan, Shantanu Srivastava           | Designing an Efficient End-to-End Machine Learning Pipeline for Real-Time Empty Shelf Detection | Proposed an end-to-end deep learning pipeline for detecting empty shelves using image data. Focuses on real-time performance, dataset preparation, and improving detection accuracy. |
| 2016 | Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi | You Only Look Once: Unified Real-Time Object Detection                                          | Introduced YOLO, a real-time object detection system using a single neural network. Forms the foundation for modern detection models like YOLOv8.                                    |
| 2019 | Mehmet Melek, Ahmet Kose, Mustafa Tekin                    | Object Detection in Shelf Images with YOLO                                                      | Applied YOLO for detecting products in retail shelf images. Demonstrates feasibility of automated shelf monitoring using computer vision.                                            |
| 2023 | Glenn Jocher, Ayush Chaurasia, Jing Qiu                    | YOLOv8: Ultralytics YOLO                                                                        | Presented YOLOv8 with improved architecture, better accuracy, and ease of deployment for real-time applications.                                                                     |
| 2024 | Ahmad Fudholi, Riza A. Pratama, Andi Sunyoto               | YOLO-Based Small-Scaled Model for On-Shelf Availability in Retail                               | Developed lightweight YOLO-based models for detecting product availability. Achieves high accuracy with reduced computational cost.                                                  |
| 2024 | Tianyu Zhang, Yifan Zhang, et al.                          | LSR-YOLO: A Lightweight Model for Retail Product Detection                                      | Proposed an optimized YOLO-based architecture to improve speed and efficiency for real-time retail detection tasks.                                                                  |
| 2017 | Joseph Redmon, Ali Farhadi                                 | YOLO9000: Better, Faster, Stronger                                                              | Extended YOLO to detect over 9000 object categories using joint training on classification and detection datasets.                                                                   |

|      |                                                               |                                                                                |                                                                                                                                         |
|------|---------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 2020 | Alexey Bochkovskiy,<br>Chien-Yao Wang,<br>Hong-Yuan Mark Liao | YOLOv4: Optimal Speed and Accuracy of Object Detection                         | Introduced improvements like CSPDarknet53 and advanced data augmentation techniques to enhance performance and speed.                   |
| 2017 | Shaoqing Ren,<br>Kaiming He, Ross Girshick, Jian Sun          | Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks | Introduced region proposal networks integrated with CNNs for accurate and efficient object detection, widely used as a benchmark model. |

# SYSTEM ARCHITECTURE – SHELF SIGHT

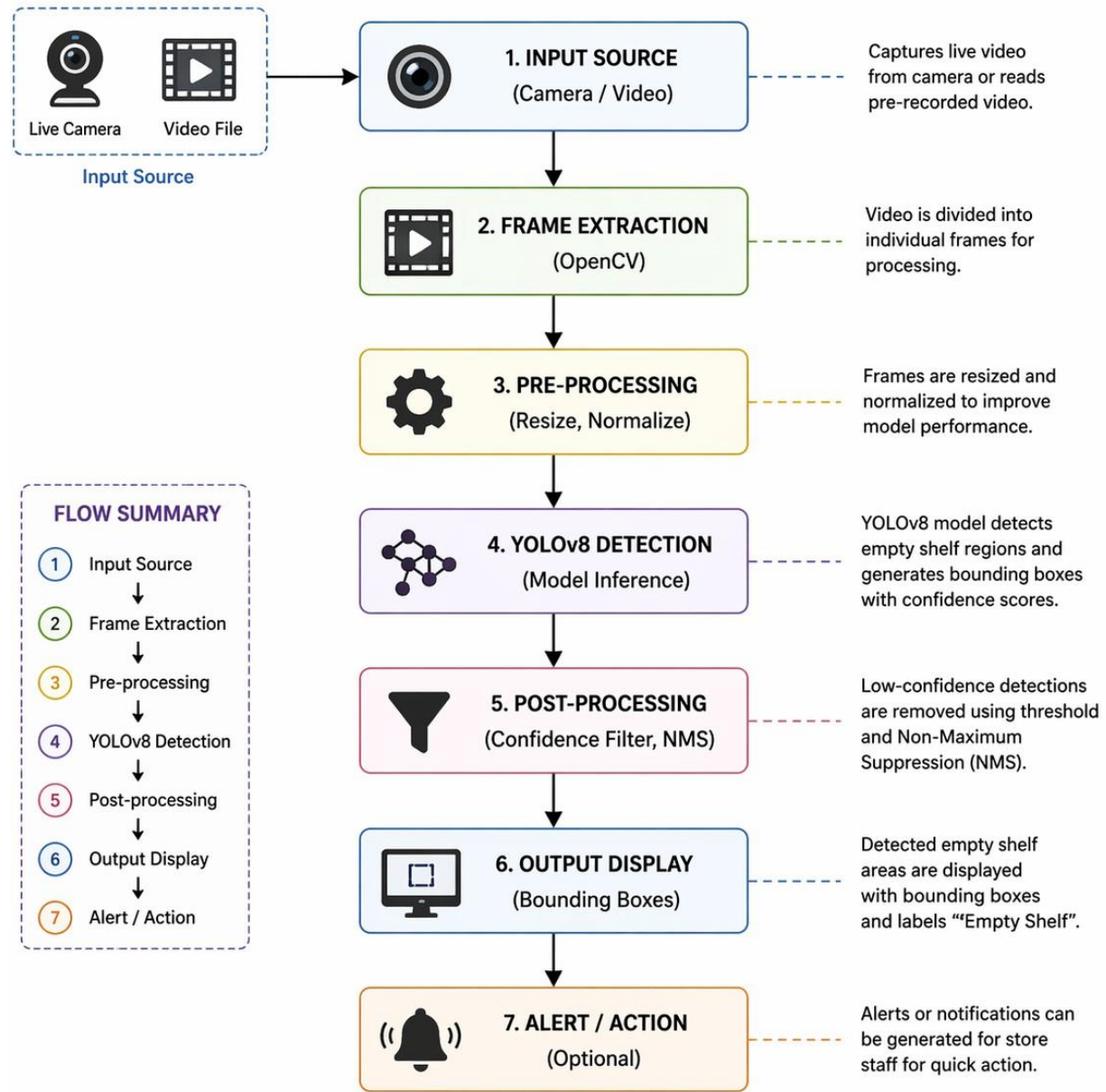


Fig. 1: System Architecture of ShelfSight

### III. PROPOSED METHODOLOGY

The proposed system, *ShelfSight*, is designed to automatically detect empty shelf spaces in retail environments using YOLOv8 and real-time video processing. The methodology is structured into multiple stages, starting from requirement analysis to deployment.

#### A. Requirement Gathering and Analysis

The first phase involves identifying system requirements based on retail store challenges. The key objective is to detect out-of-stock (OOS) situations in real time. Functional requirements include capturing video input, detecting empty shelf regions, and displaying results visually. Non-functional requirements include high detection accuracy, low latency, scalability, and ease of deployment. The system must operate efficiently under varying lighting conditions, camera angles, and shelf arrangements.

#### B. Data Collection and Preparation

A retail shelf dataset is collected from publicly available sources such as Roboflow. The dataset includes images of shelves with both filled and empty spaces under diverse real-world conditions. Images are annotated using bounding boxes to label empty shelf regions. The dataset is then split into training, validation, and testing sets to ensure proper model generalization and performance evaluation.

#### C. Model Selection

The system uses YOLOv8, a state-of-the-art object detection model known for its balance between speed and accuracy. It performs object detection in a single stage, making it highly suitable for real-time applications such as continuous shelf monitoring.

#### D. Model Training

The annotated dataset is used to train the YOLOv8 model. During training, the model learns spatial and visual patterns associated with empty shelf regions. Training involves optimization of loss functions related to bounding box regression and classification. Model performance is evaluated using metrics such as precision, recall, and mean Average Precision (mAP). Hyperparameters such as learning rate, batch size, and epochs are tuned for optimal results.

#### E. Real-Time Detection System

The trained model is integrated into a real-time detection pipeline using OpenCV. The system accepts input from live camera feeds or recorded video. Each frame is processed individually, and the YOLOv8 model detects empty shelf regions in real time.

#### F. Output Visualization

The detected empty areas are highlighted using bounding boxes and labeled as “Empty Shelf.” This visual output enables store staff to quickly identify and respond to stock shortages.

#### G. System Deployment

The system is implemented using Python and open-source libraries. It can be deployed on standard computing devices with camera support. The modular architecture allows integration with existing retail systems and supports scalability for larger store environments.

#### H. Performance Optimization

To improve efficiency, techniques such as frame resizing, frame skipping, and GPU acceleration are applied. Lightweight YOLO variants can also be used to reduce computational overhead while maintaining acceptable accuracy.

### IV. SYSTEM IMPLEMENTATION

The *ShelfSight* system is implemented to detect empty shelf spaces using video input and a trained YOLOv8 model. The system is designed to process visual data efficiently and provide real-time detection results for retail environments.

#### A. Input Module

The system accepts input in the form of a live camera feed or recorded video of retail shelves. This flexibility allows the system to be used for both real-time monitoring and offline analysis. The input data includes different shelf conditions such as fully stocked, partially filled, and empty spaces.

#### B. Frame Processing

The input video is divided into individual frames using OpenCV. Each frame is resized and pre-processed to maintain a balance between detection accuracy and computational efficiency. This step ensures that the system can process frames quickly while preserving important visual details.

#### C. Detection Module

Each processed frame is passed to the YOLOv8 model, which performs object detection in a single stage. The model analyzes spatial features in the image and identifies empty shelf regions by generating bounding boxes along with confidence scores. This enables accurate localization of empty areas.

#### D. Filtering and Optimization

To improve reliability, the system filters out low-confidence detections. This reduces false positives and ensures that only accurate empty shelf predictions are displayed. Additional optimizations such as frame resizing and efficient processing help maintain real-time performance.

#### E. Output Module

The detected empty shelf regions are displayed on the output screen using bounding boxes labeled “Empty Shelf.” The results are shown as a continuous video stream, making it easy for store staff to identify stock issues quickly.

### F. System Workflow

The overall workflow of the system follows a sequential pipeline where input video is captured, frames are processed, detections are performed, and results are displayed. This integrated pipeline ensures smooth and uninterrupted operation of the system.

## V. RESULTS AND DISCUSSION

The developed system successfully detects empty shelf spaces using the YOLOv8 model and real-time video input. The system processes both live camera feeds and recorded videos, making it suitable for practical retail environments.

During testing, the model was able to identify empty shelf regions accurately under different conditions such as varying lighting, shelf arrangements, and camera angles. The use of bounding boxes with confidence scores helps in clearly highlighting the detected empty areas.

The detection module performs efficiently in real-time, with minimal delay between frame processing and output display. This ensures continuous monitoring of shelf conditions without interruption. The filtering of low-confidence detections improves accuracy and reduces false positives.

The output results are displayed as a video stream with labeled bounding boxes indicating “Empty Shelf.” This provides a clear visual representation, allowing store staff to quickly identify and take action on out-of-stock items.

The overall system performance demonstrates that the proposed approach is effective, scalable, and suitable for real-world retail applications. The results confirm that using deep learning-based object detection can significantly improve shelf monitoring compared to manual methods.

The system outputs and detection results are illustrated in the figures below, which represent different scenarios of empty shelf detection.

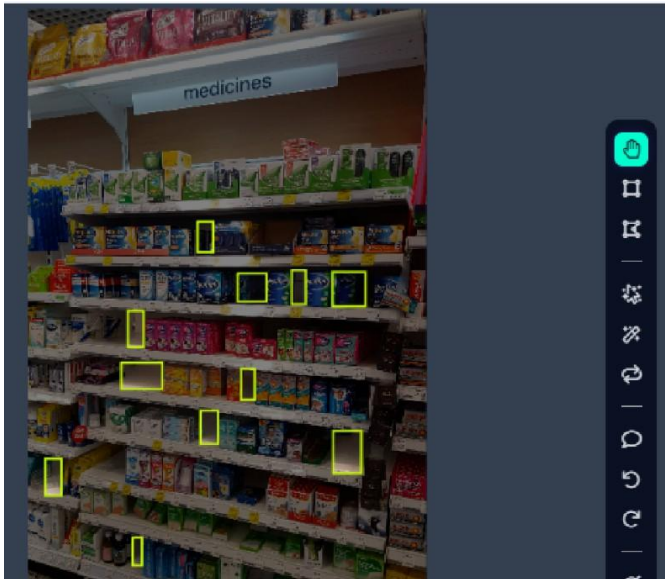


Fig. 2: Output

## VI. CONCLUSION

The proposed system, *ShelfSight*, demonstrates the effective use of deep learning for detecting empty shelf spaces in retail environments. By utilizing YOLOv8 and real-time video processing, the system identifies out-of-stock regions efficiently and accurately.

The implementation reduces dependency on manual inspection, improves response time, and enhances store management. The system performs reliably under varying conditions and provides clear visual outputs for easy interpretation.

Overall, the project shows that computer vision techniques can effectively address real-world challenges in the retail sector. The system is scalable and can be extended with features such as alert mechanisms, inventory integration, and data analytics for further improvement.

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